

Ultimate Nutrition

Glucosamine Chondroitin MSM

Glucosamine, chondroitin sulfate and MSM are substances found naturally in the body. Glucosamine is a form of amino sugar that is believed to play a role in cartilage formation and repair. Chondroitin sulfate is part of a large protein molecule (proteoglycan) that gives cartilage elasticity. MSM is a source of biological sulfur, which is a major component in many of the body's proteins, tissues, hormones and enzymes.

Increased activity/exercise, damages to the body and degenerative diseases may overcome the rate at which they are produced and used affecting the production of proteoglycans. This in turn decreases synovial fluid that cushions and prevents the joints from rubbing together. Cartilage can begin to deteriorate and weaken (less collagen) producing damage, stiffness and pain.

Osteoarthritis is associated with a breakdown of cartilage in joints and commonly occurs in the hips, knees and spine. Osteoarthritis occurs when there is significant wearing of the articular cartilage of the joint. Articular cartilage is the cushioning substance on the ends of bones and consists of glycosaminoglycans. This substance is very viscoelastic and works well as a cushion. As the articular cartilage is worn down, swelling and pain occur due to inflammation.

Chondroitin, glucosamine and MSM, as dietary supplements, are used together to help ease the symptoms of osteoarthritis. Glucosamine is an amino sugar normally produced in the human body. It is necessary for the construction and repair of connective tissue and healthy joint cartilage. Cartilage is the protein part of the fibrous substance that holds the joints. It is also used in the formation of tendons, membranes, bones and other tissues.

Glucosamine is a major building block of proteoglycans and is needed to make glycosaminoglycans: proteins that bind water in the cartilage matrix. Regular intake of glucosamine has been shown to slow Osteoarthritis progression by halting joint space narrowing and so maintain the health and mobility of the affected joints.

Chondroitin sulfate also occurs naturally in the body. Like glucosamine, chondroitin is a building block of proteoglycans: large protein molecules that give cartilage elasticity. Chondroitin sulfate acts by attracting fluid into the proteoglycan molecules and stimulates collagen production, an ingredient that forms part of the cushioning structure at the ends of the joints. Oral administration of chondroitin also prompts a significant increase in hyaluronic acid, a major component of synovial fluid: the lubricating fluid in joints.

Methyl Sulfonyl Methane (MSM) is a nutritional form of sulfur. MSM acts as a transport molecule for elemental sulfur, which is required for the amino acids: methionine and cysteine. Amino acids are the building blocks of protein. Proteins are required for growth and repair and are involved in; the formation of healthy connective tissues, joint function,

proper enzyme activity and hormone balance, along with the proper function of the immune system. MSM has very important roles in the body because of its close relationship with proteins.

MSM is useful for relieving inflammation and pain in joints that are associated with osteoarthritis, rheumatoid arthritis, muscle pain, fatigue and back pain. MSM relieves inflammation and pain by helping to equalize cell pressure.

SELECTED REFERENCES:

Chou MM, Vergnolle N, McDougall JJ, Wallace JL, Marty S, Teskey V, Buret AG. Effects of chondroitin and glucosamine sulfate in a dietary bar formulation on inflammation, interleukin-1beta, matrix metalloprotease-9, and cartilage damage in arthritis. *Exp Biol Med* (Maywood). 2005 Apr;230(4):255-62.

Hathcock JN, Shao A. Risk assessment for glucosamine and chondroitin sulfate. *Regul Toxicol Pharmacol*. 2007 Feb;47(1):78-83. Epub 2006 Aug 30.

Kim LS, Axelrod LJ, Howard P et al. Efficacy of methylsulfonylmethane (MSM) in osteoarthritis pain of the knee: a pilot clinical trial. *Osteoarthritis Cartilage*. 2006 Mar;14(3):286-94. Epub 2005 Nov 23.

Reginster JY, Deroisy R, Rovati LC, Lee RL, Lejeune E, Bruyere O, Giacovelli G, Henrotin Y, Dacre JE, Gossett C. Long-term effects of glucosamine sulphate on osteoarthritis progression: a randomised, placebo-controlled clinical trial. *Lancet*. 2001 Jan 27;357(9252):251-6.

Usha P.R., Naidu M.U.R. Randomised, double-blind, parallel, placebo-controlled study of oral glucosamine, methylsulfonylmethane and their combination in osteoarthritis. *Clinical Drug Investigation*. 24(6):353-363, 2004.

FREQUENTLY ASKED QUESTIONS:

1: Who should take Glucosamine Chondroitin MSM?

It is recommended for anyone with uncomfortable, stiff and aching joints. It can also benefit those who participate in regular physical activity by helping to maintain joint health and reduce stress from repetitive or jarring motions.

2: Is there anyone who shouldn't take this product?

If you are allergic to shellfish or fish, consult your doctor before deciding to take Glucosamine Chondroitin MSM.

3: How long will it take for me to feel positive results?

The benefits of Glucosamine Chondroitin MSM supplementation develop over time. Generally, it may take 2 to 8 weeks before you see noticeable results.

4: What is MSM and why is it recommended?

MSM is a natural sulfur-containing compound that is produced in the human body.

Although MSM occurs naturally in foods, even moderate food processing destroys most of it. It provides an important source of sulfur, which plants, animals and humans need for a wide variety of processes.